

Sai Nishchal Gamini

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Education

Master's Digital Engineering (Computer Science)

Otto von Guericke University

2024/04 – present | Magdeburg, Germany

Bachelor's Mechanical Engineering, SRKR Engineering College

2017/07 – 2021/07 | Bhimavaram, India

Projects

Multimodal RAG-Based Conversational AI

- Developed an AI-powered conversational system using Python, LLMs, and Retrieval-Augmented Generation (RAG) for intelligent document understanding.
- Built the chatbot frontend using React and integrated it with backend services through RESTful APIs.
- Designed scalable retrieval pipelines using embeddings, semantic search, vector databases (Milvus), and graph-based reranking.
- Developed modular backend components using LangChain and Hugging Face Transformers for efficient information retrieval.

Explainable Deep Learning for Image Classification

- Developed and trained deep learning models using PyTorch for image classification.
- Executed the complete machine learning workflow in a Docker environment, ensuring reproducible development and deployment.
- Applied Explainable AI (XAI) techniques, to improve model interpretability.
- Performed model evaluation and hyperparameter tuning through structured experimentation.

Community Interaction Platform (Full-Stack Web Application)

- Developed a full-stack web application using React, TypeScript, Node.js, MongoDB, and RESTful APIs.
- Built reusable, responsive React components for content sharing and community engagement using a component-based architecture.
- Integrated frontend components with backend RESTful APIs for authentication, user management, and data retrieval.
- Applied React Hooks, modular software design, and Git-based version control to develop clean and maintainable software.

Full-Stack Ticket Booking Application (Infosys Project)

- Developed a full-stack ticket booking system using React, TypeScript, Java, Spring Boot, RESTful APIs, Maven, and SQL.
- Designed responsive frontend interfaces for booking workflows and integrated them with Spring Boot RESTful APIs.
- Developed backend services for authentication, booking management, and transaction processing using a modular architecture.
- Designed and implemented a relational SQL database schema while following clean code principles and Git-based version control.

Data Analysis and Visualization of Persistent Memory Systems (Intel Optane PMem)

- Conducted exploratory data analysis (EDA) and scientific analysis on a dataset of 261 research publications.
- Performed data preparation, processing, and visualization using Python to identify trends, benchmarks, and research insights.
- Applied statistical analysis and model evaluation techniques to extract meaningful patterns from structured and unstructured data.
- Communicated findings through structured reports and visual analytics.

Skills

Artificial Engineering & Generative AI

- Large Language Models (LLMs)
- Retrieval-Augmented Generation (RAG)
- AI Agents
- Natural Language Processing (NLP)
- Knowledge Graphs
- Semantic Search
- Information Extraction
- Embeddings & Vector Representations
- Multimodal AI
- Machine Learning
- Deep Learning

Development Tools

- Git & GitHub
- Docker
- Kubernetes
- Maven
- Linux

Programming Languages

- Python
- C++

Information Retrieval & Knowledge Systems

- Retrieval-Augmented Generation (RAG)
- Vector Search
- Knowledge Graphs
- Information Retrieval
- Document Processing
- Document Analysis
- Unstructured Data Processing

Frontend development

- React
- React Hooks
- HTML/CSS
- Responsive Web Design
- RESTful API Integration

AI Frameworks & Libraries

- LangChain
- Hugging Face Transformers
- PyTorch
- Scikit-learn
- NumPy
- Pandas
- Milvus

Software Engineering

- Git
- Docker
- Linux
- REST APIs
- Flask
- SQL

Backend development

- RESTful APIs
- Responsive Web Design
- React Hooks
- MongoDB

Cloud Platform

- Google Cloud Platform (GCP)
- Kubernetes
- Terraform
- Linux
- Cloud Infrastructure Concepts
- AWS (Foundational Knowledge / Learning)

Papers

Conference on Computational Linguistics (CLiC-it 2025) [!\[\]\(291e070cef6c4d5e78fefe4696ef53be_img.jpg\)](#) ,
Diffusion-Aided RAG: Elevating Dense-Retrieval Chatbots via Graph-Based Diffusion Reranking

Languages

- English(C1)
- German(B1)